

Autonomous Underwater Vehicle to Quantify Ocean Water Pollution

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E80: Experimental Engineering, SP26



Problem Statement

This Experimental Engineering final project uses an autonomous underwater vehicle aims to investigate the levels of water pollution at Dana Point as a result of human activity and increasing carbon emissions. This project utilizes a combination of 4 sensors to autonomously collect data and quantify levels of water pollution at Dana Point. These four sensors are **pH**, **temperature**, **turbidity**, and **pressure**.

Impact

The impact of this project confronts the global issue of rising carbon emissions as a result of human industrial activity. As carbon emissions in the atmosphere increase, the ocean begins absorbing excess carbon dioxide as well, contributing to ocean acidification and other sources of pollution. This project aims to quantify the levels of ocean water pollution through 3 properties: turbidity, pH, and temperature.

Future Work

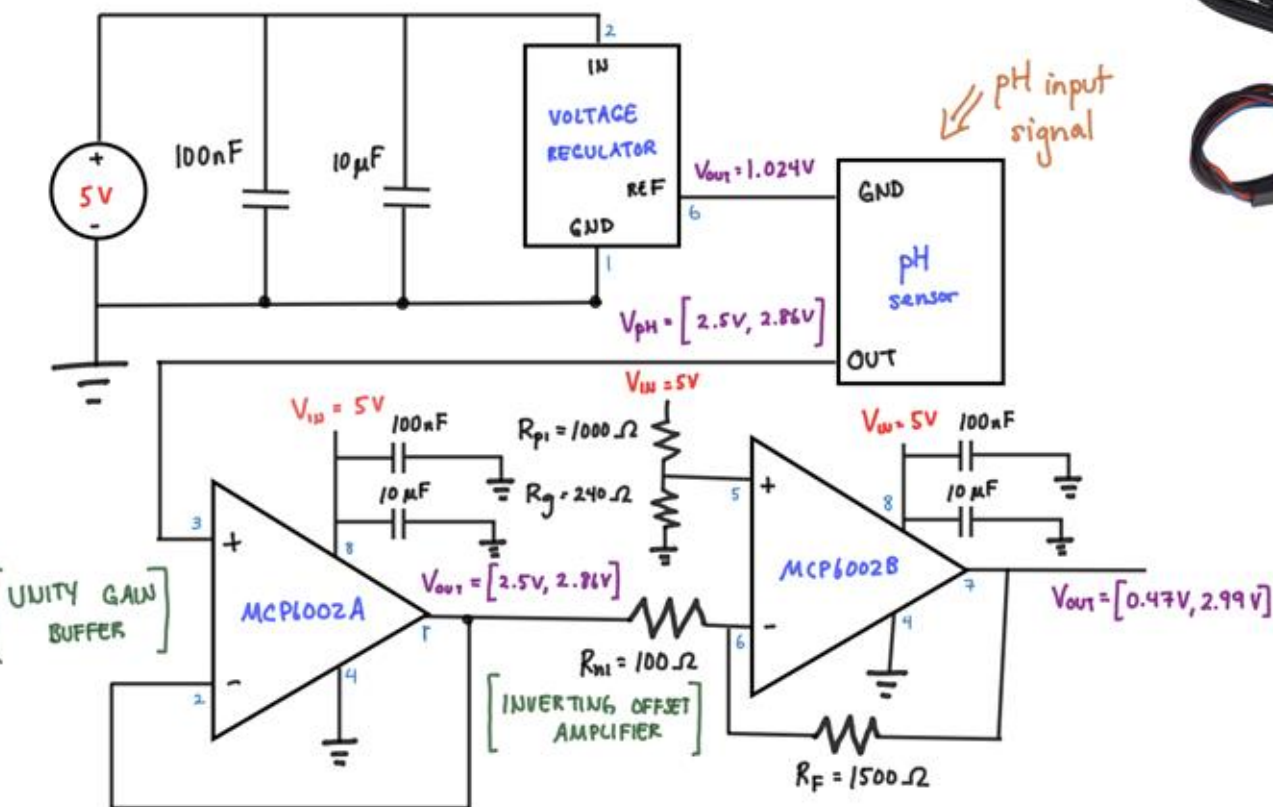
Future work on this project could include developing methods to reduce noise from seawater conductance and motor currents. Globally, this project could be expanded to develop pollution sensor stations that provide real-time data on ocean pollution levels. Ultimately, the team hopes to contribute towards creating a system of equitable information distribution that allows people to make informed decisions about their health.

Autonomous Underwater Vehicle (AUV) Features

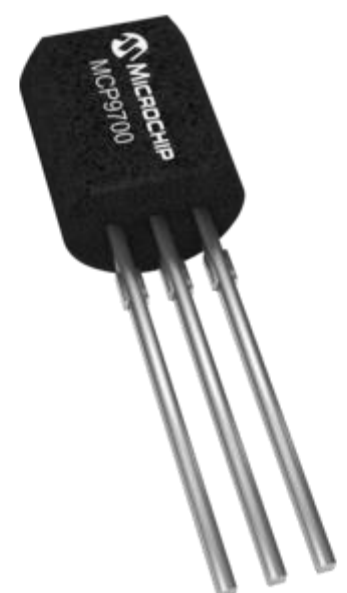
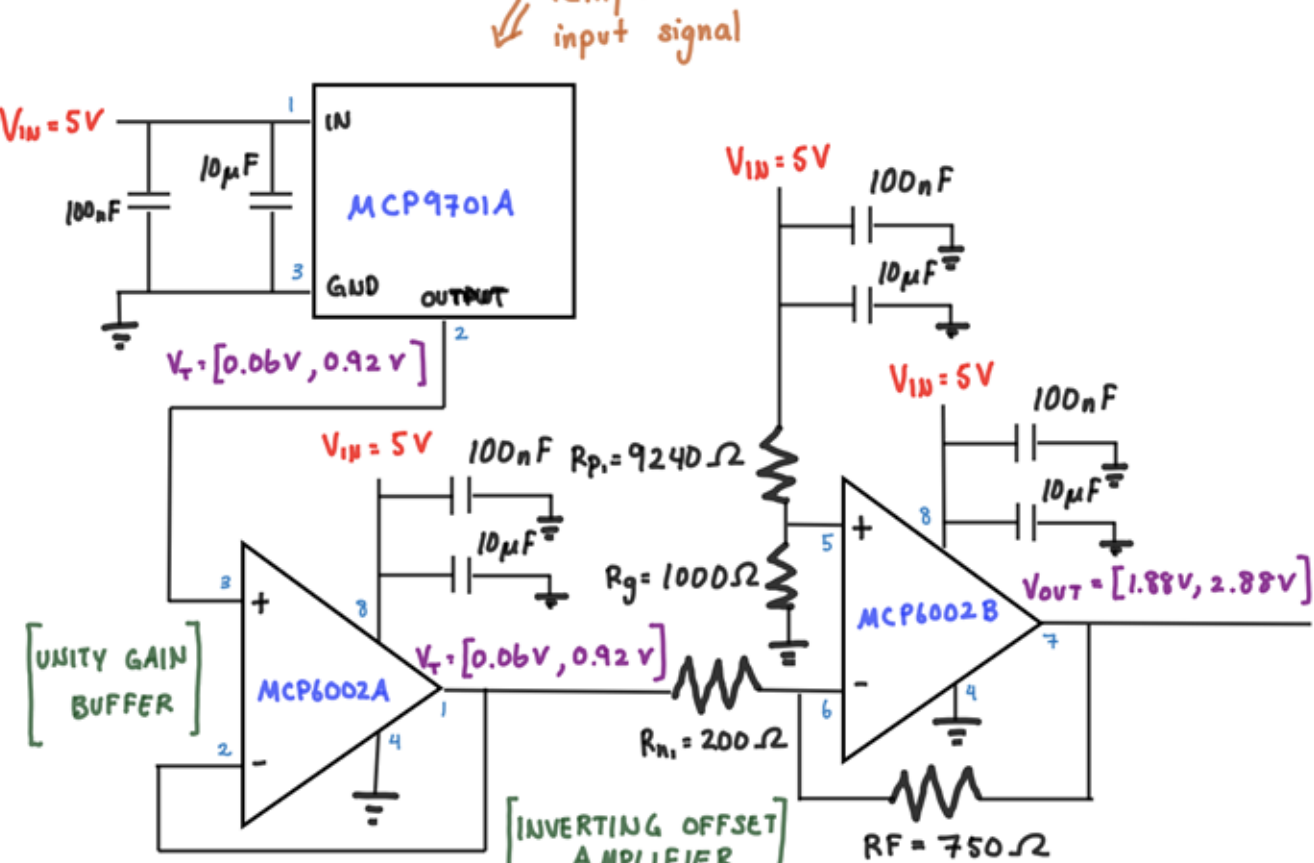
Electrical Schematic & Sensors

Mechanical Design

pH sensor

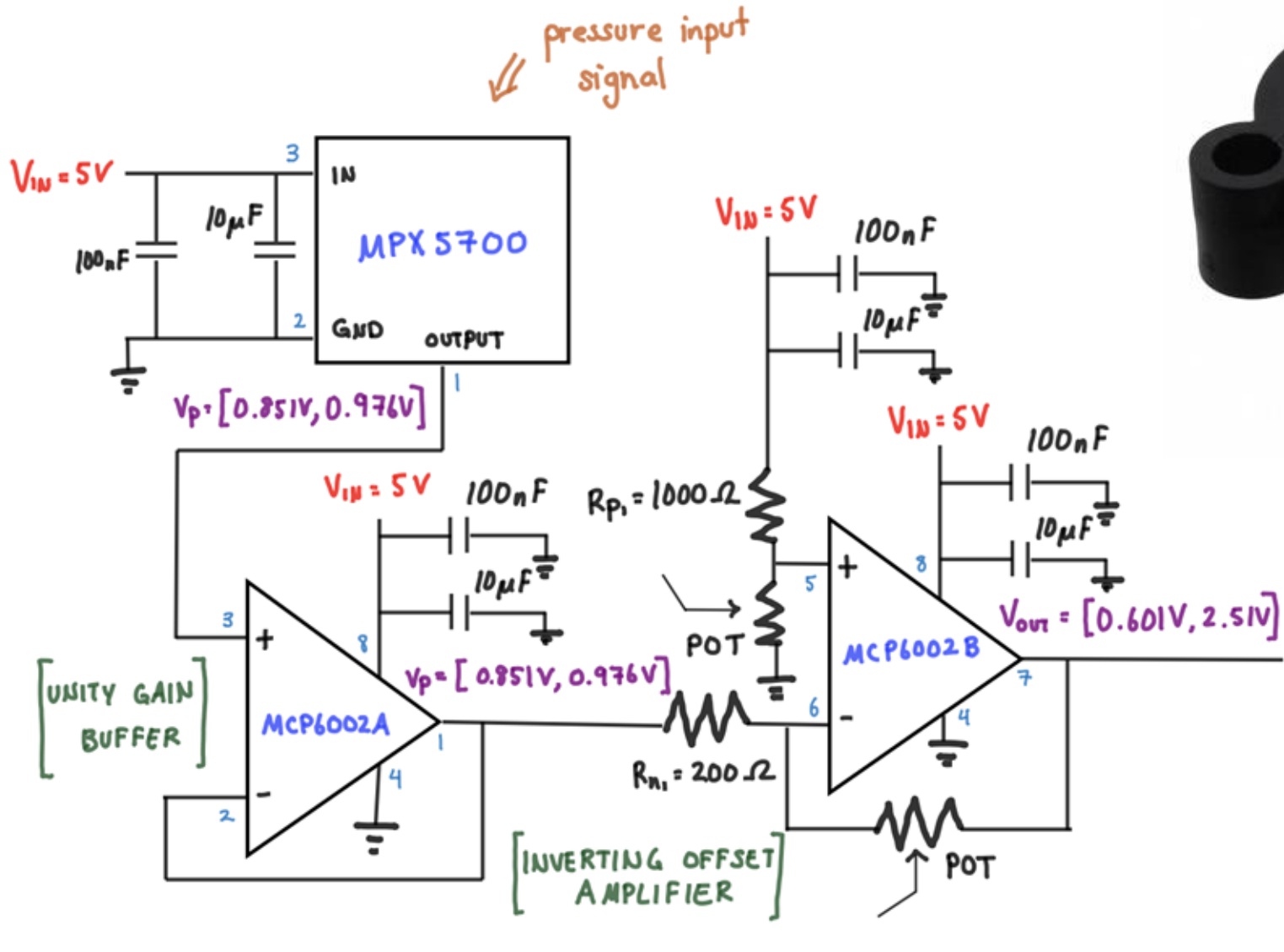
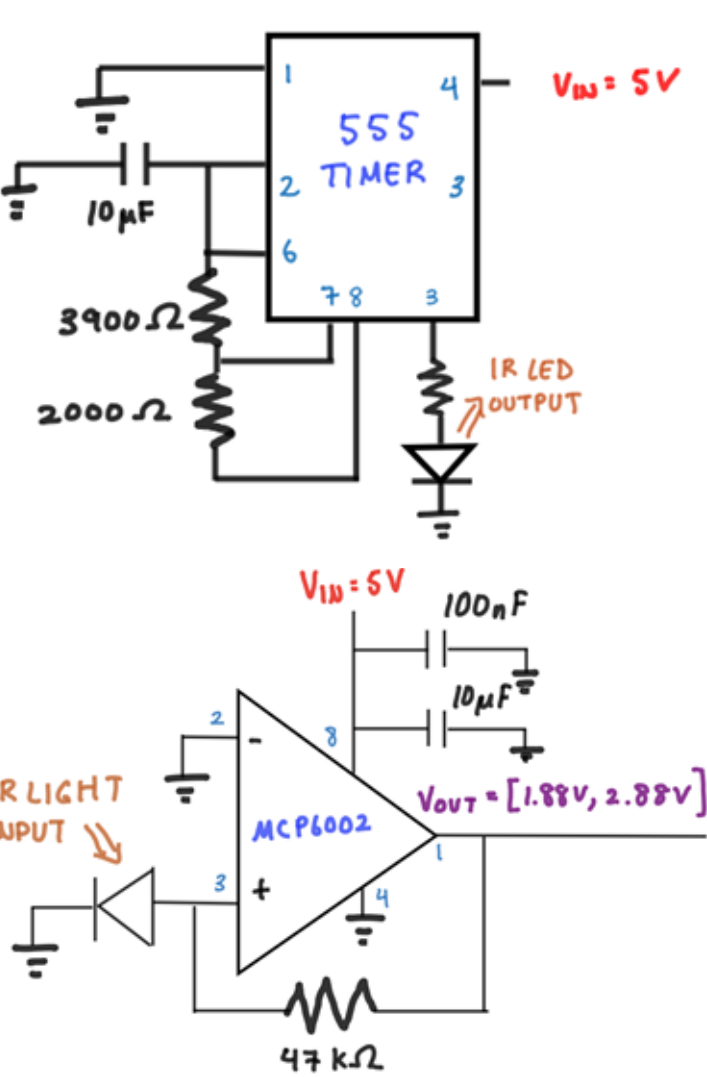


Temperature sensor

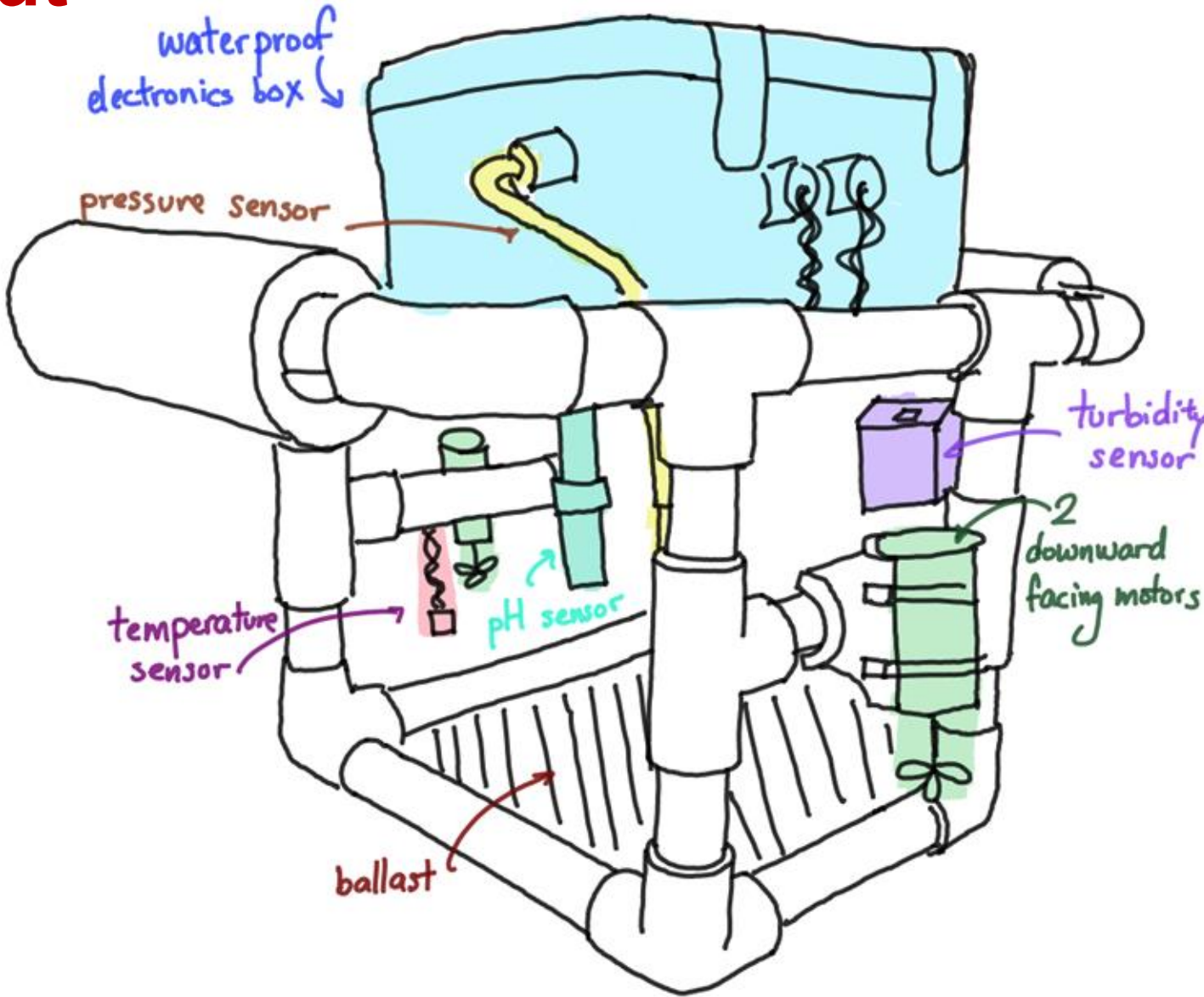


Pressure sensor

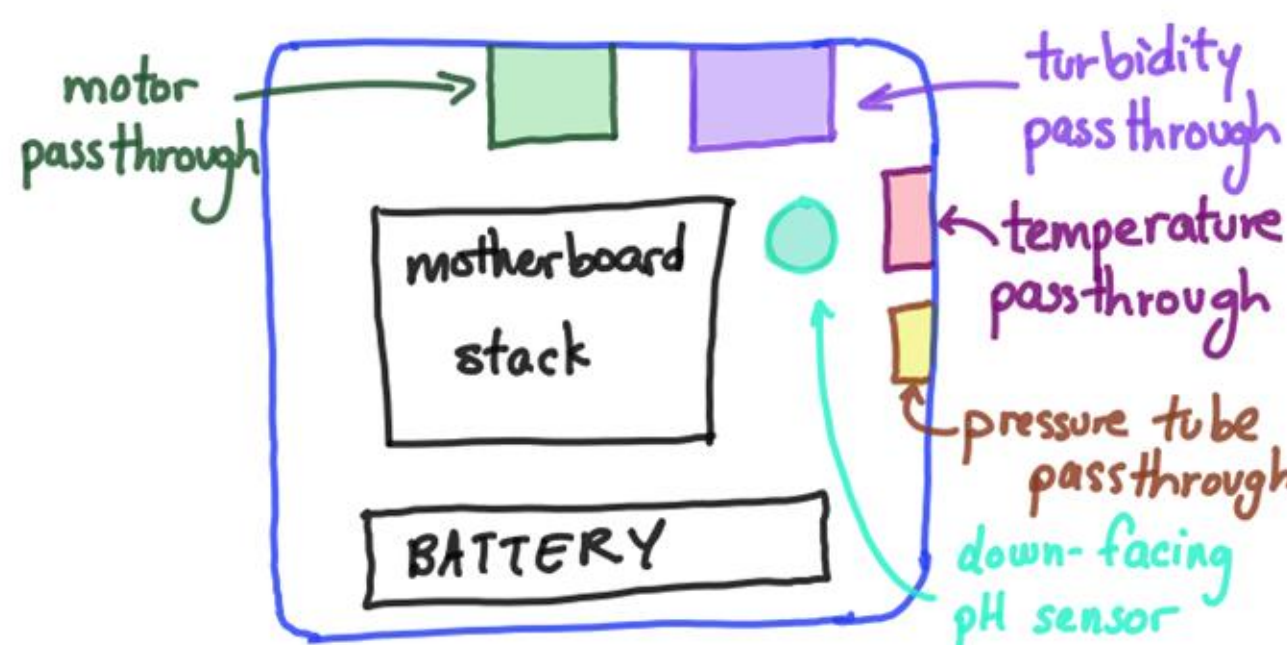
Turbidity sensor



Robot layout



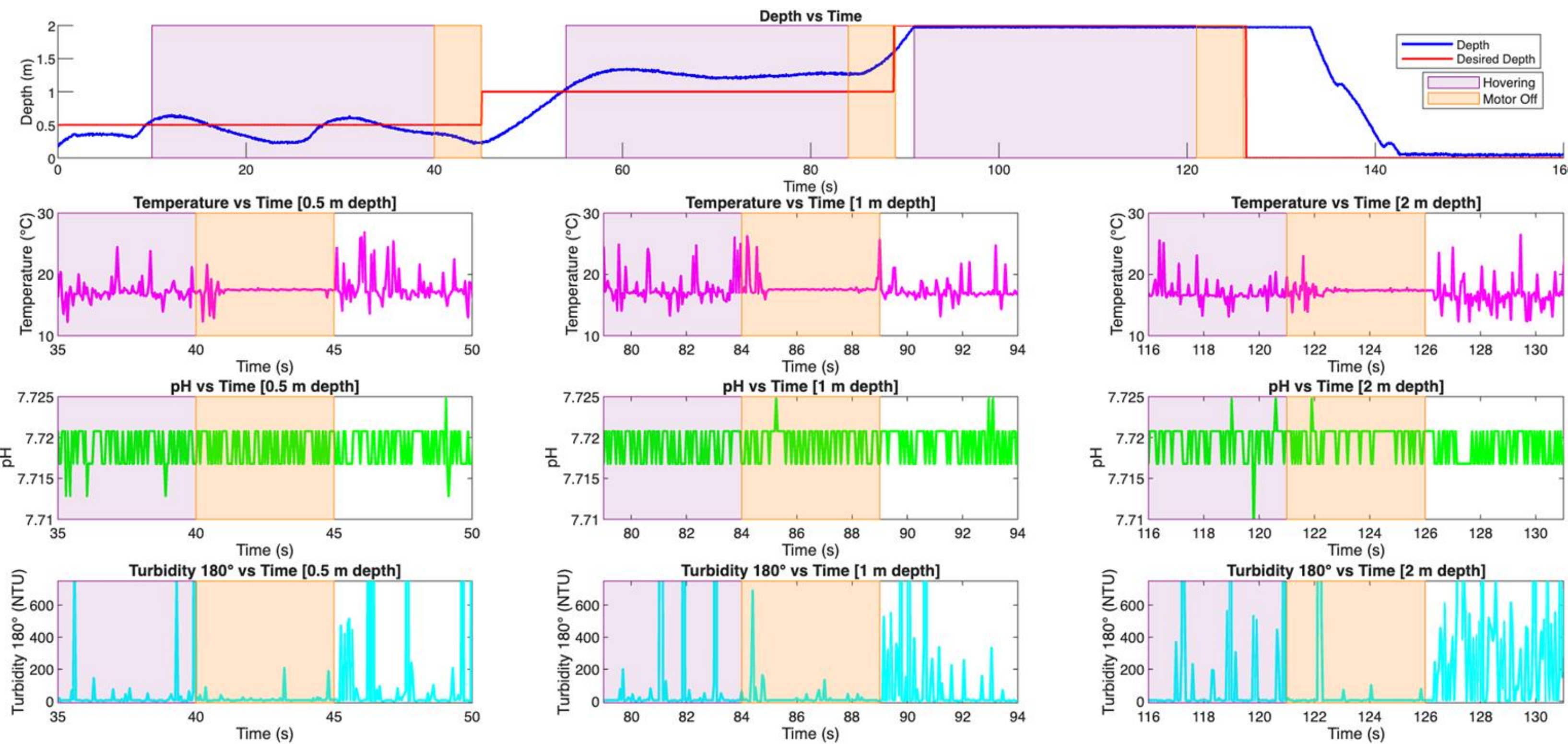
Electronics box layout



Experimental Data and Results

Results & Conclusions

Deployment Details



Dana Point Data Conclusions

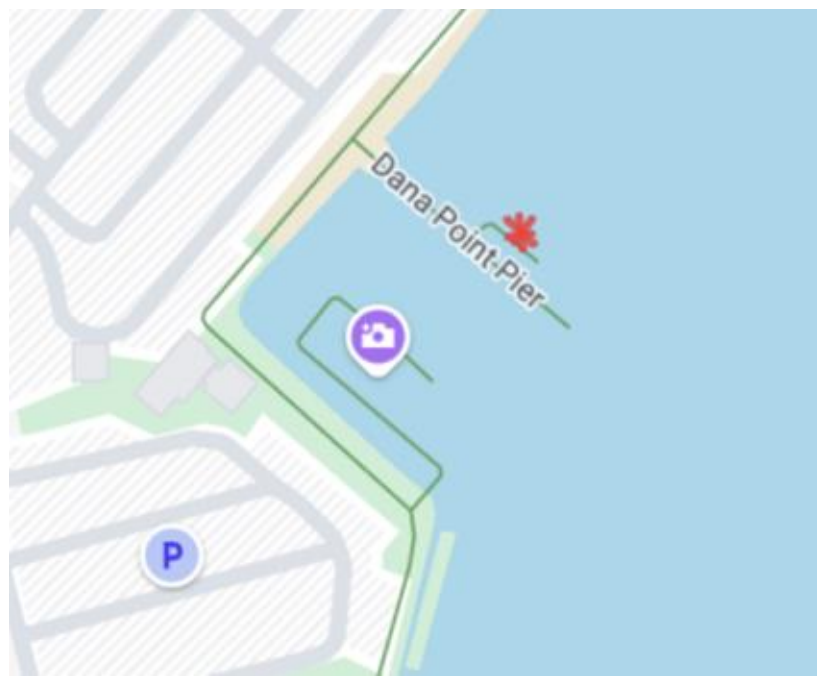
The team observed that pH remained constant, which aligns with the team's initial expectations. Temperature is expected to decrease as the robot travels down, which was not reflected in the data collected. There is a large increase in turbidity, likely a result of increased sediment concentration towards the bottom of the ocean. Sensor values were compared to ground truth values to ensure sensor readings were reasonable.

Location(s): Bernard Field Station (BFS), Dana Point Pier (33.462300, -117.705964)

Testing Protocol: At each waypoint, the robot had a 30-second hovering period for sensor acclimation and a 5-second motor off period to reduce signal noise induced by motor currents. The robot battery was disconnected anytime a team member manipulated the circuits to ensure safety.

Dana Point Deployment Location

BFS Deployment Location



	0.5 m depth	1 m depth	2 m depth
Temperature (°C)	17.34	17.86	17.33
pH	7.72	7.72	7.72
Turbidity (NTU)	12.72	21.97	110.83

Sensor Calibration

Pressure sensor was calibrated in the tank room for up to 1 meter of depth, and then extrapolated to the full 2 meter range. **Temperature sensor** was calibrated in lab with water baths ranging from 14°C to 24°C. **pH sensor** was calibrated in buffer solutions ranging from pH 6 to pH 9. **Turbidity sensor** was calibrated in lab using a water and milk powder solution for NTU values ranging from 0 NTU to 100 NTU. All sensors were calibrated to match the expected range of values collected at Dana Point.

Ground Truth

	Dana Point
Temperature	18.2°C (collected at surface)
pH	7.9 (collected at surface)
Depth	0.5 m increments on tether
Turbidity	23.25 NTU (collected at surface)

Acknowledgements

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